

# Resolution of the Pál–Vértesi conjecture: the $I_{3322}$ quantum value is attained spatially but not in finite dimension

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## Abstract

Let  $S = \omega_{\text{tensor}}(I_{3322}) = \omega_{\text{commuting}}(I_{3322})$  denote the quantum supremum of the  $I_{3322}$  Bell functional in the Collins–Gisin normalization (classical bound 0; qubit and qutrit maximum exactly  $1/4$ ). We report a computer-assisted, exactly certified resolution of the attainment questions for  $S$ : (i)  $S$  lies in the certified window  $(0.2508753845015185, 0.250875388108398]$ , and the tensor-product and commuting-operator suprema coincide at one common value; (ii) **no finite-dimensional quantum strategy attains  $S$**  — any dimension, pure or mixed states, projective or POVM measurements — resolving the conjecture of Pál and Vértesi (2010); (iii)  $S$  **is** attained by a normal spatial strategy on  $\ell^2(\mathbb{Z}) \otimes \ell^2(\mathbb{Z})$ . Consequently  $C_q(3, 3; 2, 2)$  is not closed — the smallest scenario by input count in which nonclosure is known — and  $C_{qs}(3, 3; 2, 2) \setminus C_q(3, 3; 2, 2) \neq \emptyset$ . The nonattainment proof runs through a canonical critical Bellman storage (concave, obtained as a greatest feasible fixed point), exact rational endpoint-exclusion certificates, a reflection-gluing inequality, and a convex-envelope theorem excluding contact plateaus; finiteness then forces the two equality transports of an exact maximizer to coincide, which caps its value at  $1/4 < S$ . The attainment proof disintegrates the scalar spectral measure of a maximizing commuting state over the orbits of its response translation and reads off an  $\ell^2$  Jacobi eigenvector, with normalizability inherited from the probability measure rather than imposed. Every load-bearing step was audited in refutation-first adversarial review whose complete records ship with the certificates, together with a full correction history: an earlier claimed proof by this program was publicly decertified after an exact audit, and nothing decertified re-enters the present chain. We discuss a structural dichotomy suggested by the results: across the known Bell-functional zoo, finite-dimensional attainment coincides with closed (finitely recurrent) optimal carriers, while every known transport-class nonattainment is carried by an infinite open chain.

## 1 Introduction

The  $I_{3322}$  inequality [1, 2] is the best-known bipartite Bell functional beyond the Clauser–Horne–Shimony–Holt family: three binary measurements per party, and a facet of the local polytope inequivalent to CHSH. In 2010, Pál and Vértesi [3] computed its quantum value numerically to high precision, exhibited a family of finite-dimensional strategies converging to  $0.2508753845\dots$ , observed that qubit strategies reach exactly  $1/4$  and that local dimension at least 12 is needed to exceed it, and conjectured that *no finite dimension suffices*: the supremum should be attained only in infinite dimension. The conjecture has stood since. Coladangelo and Stark, in constructing the first explicit bipartite correlation attainable only in infinite dimension (on larger

alphabets:  $4 \times 5$  inputs, 3 outputs), wrote of the  $I_{3322}$  case that “an analytical proof has remained elusive” [4].

This paper reports that resolution, in both directions, by computer-assisted proof with exact certificates:

**Theorem 1** (Nonattainment). *In the Collins–Gisin normalization, let  $S = \omega_{\text{tensor}}(I_{3322}) = \omega_{\text{commuting}}(I_{3322})$  be the quantum supremum, certified to lie in  $(0.2508753845015185, 0.250875388108398]$ . Then no finite-dimensional quantum strategy attains  $S$ : for every finite dimension  $d$ , every (pure or mixed) state on  $\mathbb{C}^d \otimes \mathbb{C}^d$  and every triple of binary measurements per party (projective or POVM),  $\langle I_{3322} \rangle < S$ .*

**Theorem 2** (Spatial attainment).  *$S$  is attained by a normal spatial strategy: there is a unit vector  $\psi_S \in \ell^2(\mathbb{Z}) \otimes \ell^2(\mathbb{Z})$  and six projective binary measurements built from the alternating Pál–Vértesi projector blocks with  $\langle \psi_S, \mathcal{B}_{3322} \psi_S \rangle = S$ .*

**Corollary 3.**  *$C_q(3, 3; 2, 2)$  is not closed, and  $C_{qs}(3, 3; 2, 2) \setminus C_q(3, 3; 2, 2) \neq \emptyset$ .*

By input count,  $(3, 3; 2, 2)$  is the smallest scenario in which nonclosure of the finite-dimensional quantum correlation set is known: the previously smallest examples are the  $(5, 5; 2, 2)$  synchronous construction of Dykema–Paulsen–Prakash [5], the  $(4, 4; 2, 2)$  and  $(4, 4; 3, 3)$  separations of Beigi [6], the  $(4, 5; 3, 3)$  correlation of Coladangelo–Stark [4], and Slofstra’s original  $184 \times 235$ -input games [7, 8]. (In the  $(2, 2; 2, 2)$  scenario every functional attains its maximum on two qubits by Jordan’s lemma [9], so three inputs is also the smallest possible venue for this phenomenon.)

## Correction history, stated first

This program previously released a claimed proof of the  $I_{3322}$  headline (exact optimum, spatial attainment, nonattainment) whose load-bearing global amplitude datum was subsequently *refuted by our own post-release exact audit*: the amplitude-compatibility equation of the original bi-infinite wall construction fails by an exactly certified margin (the mismatch lies in  $[1.40 \times 10^{-4}, 1.79 \times 10^{-4}]$ , excluding zero). The claims were publicly decertified within a day, and the repository’s release history preserves the full correction record. The results announced here were then rebuilt on independent routes; the decertified fixed point, amplitude profile, and tail constants appear on explicit *not-used* lists in the certificate dependency records, and the failed compatibility equation is not repaired anywhere in the present chain — the new attainment route never poses it (Section 4). Every load-bearing theorem below then survived three rounds (Theorem 1) plus one round (Theorem 2) of independent, refutation-first adversarial review; the signed final review records ship with the certificates [23]. We state this history first because computer-assisted claims are only as credible as their failure handling; the correction record is part of the result.

## 2 Setting and certified inputs

We use the Collins–Gisin normalization of  $I_{3322}$  [2, 3]: the classical (local deterministic) bound is 0; the maximum over two-qubit — indeed, by [3], over qubit and qutrit — strategies is exactly  $1/4$ ; Pál–Vértesi report that local dimension  $\geq 12$  is required to exceed  $1/4$ , with their strategy family converging to  $0.250875384514\dots$  [3]. Throughout,  $S$  denotes the common quantum value of Theorem 1, identified *only* by its certified window; we do not claim  $S$  equals the historical

decimal, nor that the Pál–Vértesi family is optimal, though all certified data are consistent with both.

The proofs consume the following exactly certified inputs, established in the public repository [23] (certificate directories cited in Section 6): (T0) a Bellman/path variational theorem giving  $\omega_{\text{tensor}} = \omega_{\text{commuting}} = S$ , together with, for every level  $q > S$ , a terminal-pivot storage  $g_q$  — the infimum of terminal Schur pivots over finite histories — which is positive, Bellman-feasible, and equicontinuous in  $q$ ; (LB) an exact 255-dimensional strategy certifying  $S > 0.2508753845015185 > 1/4$  (rational square-root floors, independently reconstructed in 160-digit interval arithmetic); (UB) an exact rational 25,601-knot Bellman witness certifying  $S \leq 0.250875388108398$ ; and (W) a generic operator weld: any positive Bellman-feasible storage at level  $q$  yields a positive decomposition  $qI - \mathcal{B} = R_0 + R_A + R_B$  with the local remainder structure of [23] (sprint 1197). The finite Pál–Vértesi block-to-Jacobi identity ([23], sprint 1206, §§2–4 only — the finite, endpoint-free form) supplies the bridge between Jacobi data and Bell strategies; the adversarial review independently replicated it at machine precision.

### 3 Nonattainment

We sketch the architecture; complete proofs, with every constant exact, are in the certificate documents [23], directory **theorem-N-four-receipts-at-S**.

*The canonical critical storage.* The map  $q \mapsto g_q$  is monotone: increasing the level by  $\eta$  increases every terminal Schur pivot by at least  $\eta$ , whence  $g_q \geq g_S + (q - S)$  for the decreasing limit  $g_S = \inf_{q>S} g_q$ , which is independent of the approximating sequence. Each terminal pivot is affine in the target label, so  $g_S$  is *concave*; and  $g_S$  is the *greatest* Bellman-feasible storage at level  $S$ , hence an exact fixed point of the Bellman operator. Concavity is elementary but decisive: it forces every corner of the contact intercept downward, which is what ultimately excludes contact plateaus.

*Exact endpoint exclusion.* Two-edge finite histories give exact rational margins: with  $r = 1/10$ , the endpoint predecessor lines exceed the storage uniformly by  $m_+ = \frac{23686917837403}{3008753881083980} > 0.0403$  and  $m_- = \frac{274562305945801}{4008753881083980} > 0.1978$ , for every level in the certified window. Consequently the remainder  $R_0$  has a uniform positive gap on all endpoint spectral sectors, so no exact or near-maximizing state carries endpoint mass, and  $x = \pm 1$  is never a Bellman predecessor at  $S$ . These are closed-form rational certificates; no branch geometry or reflected wing is used.

*The equality module of a hypothetical maximizer.* Suppose a finite-dimensional strategy attains  $S$ . (Purification is unnecessary:  $\text{Tr}(\rho R) = 0$  with  $R \succeq 0$  gives  $R\rho = 0$  directly; POVMs are handled by a finite Naimark dilation preserving dimension, tensor structure, and value.) Applying the weld at levels  $q_n \downarrow S$  and using  $\sum_\nu \text{Tr}(\rho R_{\nu,n}) = q_n - S \rightarrow 0$ , the endpoint gaps kill boundary mass first; on the remaining interior finite spectrum the coefficients converge, and the limiting positive remainders annihilate  $\rho$ . Thus the maximizer carries an exact critical equality module at  $S$  — *without* any claim that the storage infimum is attained (the endpoint boundary layer is bypassed, not solved).

*Zero-set geometry.* On the equality module, the scalar remainder symbol vanishes. Reflected finite histories give the gluing inequality  $g_S(x)g_S(-x) \geq b(x)^2$  with  $b(x)^2 = (1 - x^2)/4$ ; full equality of the two Bellman inequalities and the Cauchy–Schwarz step forces  $g_S(x)g_S(-x) = b(x)^2$  at every occupied pair, and a Monge (supermodularity) argument makes the full zero set monotone. The decisive step is a convex-envelope theorem: writing  $C(x) = S + 1 - \frac{x}{2} - b(x)^2/g_S(x)$  on the open interval  $(-1, 1)$  — where  $C \geq g_S(0) > 0$  by feasibility at the origin,

so the greatest convex minorant  $H$  exists — concavity of  $g_S$  makes every corner of  $C$  point downward, while a kink of  $H$  at a contact point would force an upward corner. Hence  $H$  is differentiable, no source serves two targets, and (by an exact involution  $(x, u) \mapsto (-u, -x)$  of the constraint system, which requires no symmetry of  $g_S$  itself) no target is served by two sources. The occupied support of any maximizer is therefore a strictly increasing one-to-one graph.

*Finiteness forces closure; closure caps the value.* The equality module carries two unitary response transports implementing the decreasing involutions  $a(u) = P^{-1}(-P(u))$  and  $b(u) = -u$  on the occupied support; the graph property makes both total. A finite totally ordered set admits exactly one decreasing bijection, so  $a = b$  on the support: every occupied pair has its reflected partner. The resulting amplitude-ratio elimination (multiplicity-uniform: the transports are scalar multiples of unitaries between complete fibres) yields, for an occupied pair  $(x, u)$ ,

$$S = xu - 1 + \sqrt{(b_x + b_u)^2 + \frac{(x-u)^2}{4}} \leq -t + \sqrt{t} \leq \frac{1}{4}, \quad t = 1 - xu,$$

contradicting  $S > 1/4$ . This proves Theorem 1. The nonclosure half of Corollary 3 follows by compactness of the behavior space: finite-dimensional behaviors approach value  $S$ ; a convergent subsequence has value exactly  $S$ ; closedness of  $C_q$  would give a finite-dimensional realization attaining  $S$ .

*Remark 4.* The only role of finite dimension is the single sentence “a finite ordered set has one decreasing bijection.” Everything else holds for arbitrary maximizers, which is exactly why Theorem 2 is possible.

## 4 Spatial attainment

The historical route to attainment constructed amplitudes along a bi-infinite wall and then demanded a global compatibility equation — the equation whose exact failure decertified the original release. The present route *dissolves* that obstruction rather than repairing it: no amplitude is ever constructed, and the equation is never posed.

By weak-\* compactness a maximizing state  $\Omega$  exists for the commuting model (whose value is always attained [13]; nonattainment is strictly a tensor-model phenomenon). The limit passage of Section 3 applies verbatim — no finite spectral support is needed once the transports are expressed through the exact operator blocks  $W = Y(B_3 - \frac{1}{2}I)$  and  $W_B = (A_3 - \frac{1}{2}I)V$ , which satisfy  $W^2 = b(X)^2$  and  $WX = -XW$  as projector identities (no global spectral involution is assumed; it need not exist). These yield exact Radon–Nikodym transport laws for the scalar spectral measure  $\mu$  of the commuting pair  $(X, U)$ . The fixed-point set of the response translation  $\tau = a \circ b$  carries no mass (on it the two transport densities coincide, which fires the quarter-ceiling elimination against  $S > 1/4$ ), so  $\mu$  disintegrates over the countable free orbits of  $\tau$  via an explicit Borel transversal. Choosing one conditional orbit measure and interleaving the weights of  $\{u_n\}$  and  $\{-u_n\}$  produces amplitudes  $\lambda_j = \sqrt{\mu_t(\{c_j\})}$  with, automatically,  $1 \leq \sum_j \lambda_j^2 \leq 2$ : *normalizability is inherited from the probability measure, not imposed* — this is the precise point where the historical obstruction disappears. The transport laws then *derive* the cocycle  $\lambda_{j+1}/\lambda_j = g_S(c_j)/b(c_j)$ , and the Bellman contact equality closes the Jacobi eigenvalue equation  $H\lambda = S\lambda$  exactly. The finite Pál–Vétesi block identity installs the normalized eigenpair as a spatial strategy with value  $S$ , proving Theorem 2 and, with Theorem 1, the separation in Corollary 3.

*Remark 5 (Asymmetry).* Numerical reconnaissance stably locates the active predecessor range near  $[-0.894, +0.898]$ : the critical envelope appears genuinely asymmetric, and no step above obtains the left wing by reflection. This is compatible with the  $u \mapsto -u$  symmetry that the closure argument forces on the *occupied orbit* of a hypothetical finite maximizer: envelope and orbit are different objects. The historical construction’s implicit reflection symmetry is, in retrospect, adjacent to the exact incompatibility that decertified it.

## 5 Discussion: attainment and open carriers

The two theorems exhibit a sharp mechanism: *finiteness forces the equality transports to close (return), and closure caps the value strictly below  $S$ ; the value above the closed ceiling is carried only by an infinite open chain.* It is natural to ask how general this picture is. A survey of the known zoo is suggestive. Every Bell functional with proven finite-dimensional attainment of its global supremum closes for an identifiable algebraic reason: bipartite correlator (XOR) functionals via Tsirelson’s Clifford construction [10, 11] (anticommutation is two-step closure), all  $(2, 2; 2, 2)$  functionals via Jordan’s lemma [9], chained inequalities (rank-2 Gram), fixed CGLMP functionals (attained at local dimension exactly  $d$ ; proven for  $d = 3, 4$  [12]), and the SOS-certified families — a sum-of-squares certificate being precisely a dimension-free algebraic closure of the optimum. Conversely, both known explicit transport-class nonattainment objects are infinite open ladders: the Coladangelo–Stark state is an exactly geometric ladder whose nonattainment proof is an explicit no-return orbit argument on the Schmidt spectrum [4], and the Pál–Vértesi family is a half-infinite Jacobi chain with offset  $2 \times 2$  blockings [3] — the same architecture, on smaller alphabets, now with the conjecture it motivated resolved. (Coladangelo and Stark explicitly suggested their orbit technique might apply to  $I_{3322}$ ; Theorem 1 may be read as the answer to that suggestion, reached by a transfer-operator route.) Embezzlement-based nonattainment [14, 15, 16] is carried by the log-flat ladder  $p_j \propto 1/j$  — open but null-recurrent rather than transient, matching its distinct operator-algebraic classification [17]. Group-theoretic and type-based obstructions [7, 5, 18, 19] form a third family outside this transport picture, and Slofstra’s undecidability results imply any general structural criterion must be restricted to a subclass. Within the two-outcome bipartite class, where Jordan’s lemma supplies the  $2 \times 2$  chain structure, we record the dichotomy as a conjecture: *finite-dimensional attainment coincides with finite recurrence of the critical transport.* A concrete test: the Dykema–Paulsen–Prakash  $(5, 5; 2, 2)$  nonclosure [5] is proved non-constructively; the conjecture predicts a forced-open orbit in its critical transport. We also note the marginal/correlator boundary: Tsirelson’s theorem covers correlators only, and it is exactly the marginal terms through which  $I_{3322}$  escapes it. Finally, two cautions from the literature: the commuting value is always attained [13] (so nonattainment is a tensor-side phenomenon), and non-saturation of NPA levels is not evidence of nonattainment [21, 22].

What remains open here: the exact identification of  $S$  beyond the certified window (the window’s width is  $3.61 \times 10^{-9}$ ), and the matching dimension-necessity lower bound (how fast  $S - S_d \rightarrow 0$  from below); a constructive upper profile with logarithmic dimension sufficiency is part of ongoing work.

## 6 Methods and verification

All load-bearing computations are exact (rational arithmetic or outward-rounded interval arithmetic), two-engine where numerical (independent reconstructions in separate codebases; e.g. the 255-dimensional lower strategy is verified in exact rationals and reproduced at 160 digits), and shipped as machine-checkable certificates in the public repository [23], with the complete correction record. The proof texts themselves were audited by independent refutation-first adversarial review: three successive rounds for Theorem 1 (the first round *refuted* an earlier restoration attempt with an explicit countermodel; the third signed the theorem with a binding claim boundary that this paper follows), and a separate round for Theorem 2 (which located and repaired two genuine gaps — a circular fixed-point exclusion and a globally nonexistent spectral involution — before signing). The complete signed verdicts ship in the certificate directories. Package guard scripts are algebraic and exact-arithmetic checks only and are labeled as such; the analytic steps are established in the reviewed documents, not by the scripts.

### Data availability

Repository: <https://github.com/Apsiape/i3322-exact-wall> (release v3.2.0). Concept DOI: [10.5281/zenodo.21782008](https://doi.org/10.5281/zenodo.21782008). Certificate directories: `certificate/production/theorem-N-four-receipts-at-S/` and `certificate/production/theorem-S-spatial-attainment-at-S/`, each including the full review records.

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